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KCMamba: Kalman-Guided Selective State-Space Model for Robust Time Series Forecasting

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Declaration

I, **András Kovács**, student of Computer Science BSc at Eötvös Loránd University, hereby declare that the work presented in this thesis was carried out independently, using only the sources listed in the bibliography. Every part taken verbatim or paraphrased from another source has been clearly marked and properly cited.

Budapest, April 2026

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Abstract

This thesis presents **KCMamba** (Kalman Core-Attention Mamba), a hybrid neural architecture that integrates a differentiable Kalman filtering mechanism into the Mamba selective state-space model (SSM) for noise-robust time series forecasting. The proposed architecture employs learned Q_{net} , R_{net} , and K_{net} subnetworks that adaptively regulate the ratio of process and measurement noise — enabling the model to behave as a passive predictor on clean data and as an active filter under noisy conditions.

We conduct empirical evaluation on three public LTSF benchmarks (Exchange-Rate, ETTh1, ETTh2) under synthetically injected noise ($\sigma \in \{0, 1, 3, 5\}$) and low data-availability regimes (stride $\in \{1, 16\}$). Our results show that KCMamba achieves an average MSE degradation of +1 528% as noise increases from $\sigma = 0$ to $\sigma = 5$, compared to +28 234% for Fair Mamba — a $18.5\times$ robustness advantage.

The thesis also describes the complete production software system: an asynchronous, event-driven trading engine built in a hexagonal architecture, a web-based dashboard, a backtesting framework, and a live Binance trading adapter.

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Chapter 1

Introduction

Time series forecasting is one of the most studied areas of applied machine learning, partly because real-world data is almost always noisy. Sensor dropouts, aggregation artefacts, and market microstructure noise all degrade the signal-to-noise ratio, and models frequently struggle with this. In the long-term forecasting (LTSF) setting the problem is harder still: given a lookback window of length L , we must predict the next H steps, where H can exceed L [1].

1.1 Motivation

Transformer-based models [1, 2] and the Mamba SSM [3, 4] address data-quality issues through implicit generalisation; they rely on the model to learn around the noise. This works on clean data, but there is no guarantee that the robustness transfers to noisy regimes.

The Kalman filter [5] takes a different route: it builds an *explicit* model of process noise (Q) and measurement noise (R), and derives a gain K from their ratio to decide how much to trust the current measurement versus the prior state. That inductive bias, if embedded in a neural network, may help precisely in noisy conditions.

This is the idea behind the **KCMamba** architecture presented in this thesis.

1.2 Research Questions

Three concrete questions are investigated:

1. Is KCMamba competitive on clean data against standard LTSF models?

2. How much MSE degradation do models suffer under synthetically injected noise ($\sigma \in \{0, 1, 3, 5\}$)?
3. Which architecture generalises better under scarce training data (stride $\in \{1, 16\}$)?

1.3 Contributions

- A description of the **KCMamba** architecture: learned Q_{net} , R_{net} , K_{net} sub-networks inside a differentiable Kalman filter, with a parallel prefix-scan GPU implementation.
- Measurements showing that the learned K adapts to noise level: $K \approx 0.71$ at $\sigma = 0$ (passive pass-through) and $K \approx 0.07$ at $\sigma = 5$ (strong filtering).
- Benchmark results on three datasets (Exchange-Rate, ETTh1, ETTh2): Fair Mamba degrades by +28 234% under noise; KCMamba by +1 528%.
- An open-source Python/PyTorch/CUDA implementation and an asynchronous trading system for deploying the model in live markets.

1.4 Thesis Structure

Chapter 2 reviews the Kalman filter, SSMS, and the LTSF literature. Chapter 3 describes the KCMamba architecture and its mathematical underpinnings. Chapter 4 presents the experimental setup and results. Chapter 5 covers the implemented software system. Chapter 6 draws conclusions and discusses future work.

Chapter 2

Related Work

2.1 The Kalman Filter and Its Extensions

The Kalman filter [5] is the optimal Bayesian estimator for linear dynamical systems under additive Gaussian noise. The state update is:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t(y_t - H\hat{x}_{t|t-1}), \quad (2.1)$$

where the gain K_t is determined by the ratio of prediction and measurement covariance:

$$K_t = P_{t|t-1}H^\top (HP_{t|t-1}H^\top + R)^{-1}. \quad (2.2)$$

When $R \rightarrow \infty$, $K \rightarrow 0$: the filter distrusts the measurement. When $R \rightarrow 0$, $K \rightarrow 1$: the measurement nearly overwrites the state. This simple rule is the core intuition behind KCMamba.

2.1.1 Learned Kalman Filters

In the neural Kalman literature, the matrices F, H, Q, R are parameterised by networks [6, 7]. The *Deep Kalman Filter* [6] allows non-linear transitions; the *Kalman Variational Autoencoder* [7] learns disentangled representations of latent states. KCMamba builds on these ideas but operates as a sequence forecaster rather than as an autoencoder.

2.2 State Space Models

2.2.1 S4

S4 [3] models a d -dimensional input sequence via an N -dimensional latent dynamic:

$$h'(t) = Ah(t) + Bu(t), \quad (2.3)$$

$$y(t) = Ch(t). \quad (2.4)$$

After discretisation: $h_t = \bar{A}h_{t-1} + \bar{B}u_t$, $y_t = Ch_t$, where $\bar{A} = e^{\Delta A}$.

2.2.2 Mamba

The key innovation of Mamba [4] is input-dependent selectivity: B , C , and Δ depend on the input u_t . This lets the model decide what to remember and what to forget. A hardware-efficient parallel scan achieves $O(T)$ memory and time complexity.

2.2.3 Fair Mamba

The comparison baseline, referred to as “Fair Mamba”, is a channel-wise Mamba with identical hyperparameters and training protocol as KCMamba. The “fair” label means the parameter budget and data are identical, so any difference in results comes solely from the architecture.

2.3 Long-Term Time Series Forecasting

2.3.1 Problem Formulation

Given $\mathbf{X} = (x_1, \dots, x_L) \in \mathbb{R}^{L \times d}$, the goal is to predict $\hat{\mathbf{X}} = (\hat{x}_{L+1}, \dots, \hat{x}_{L+H})$ by minimising normalised MSE.

2.3.2 Reference Models

- *Informer* reduces Transformer complexity to $O(L \log L)$ via ProbSparse attention [1].

- *Autoformer* applies series decomposition and auto-correlation-based attention [2].
- *DLinear* achieves surprisingly strong results with a plain linear projection, questioning the necessity of complex architectures [8] — a reminder that a simpler model is not always worse.

2.3.3 Datasets

- **Exchange-Rate** [9]: 8 currency exchange rates, 7 588 daily samples. Its near-unit-root dynamics make the Kalman structure particularly effective.
- **ETTh1 / ETTh2** [1]: Hourly transformer-station electricity data ($d = 7$, 17 420 steps). ETTh1 is strongly periodic; ETTh2 is noisier.
- **Weather, ECL**: Not included in our experiments but are standard benchmarks in the LTSF literature.

2.4 Noise Robustness

The noise sensitivity of time series models is rarely studied in a systematic way. PatchTST [10] indirectly improves noise tolerance through patch tokenisation, but robustness is not its primary goal. Augmentation-based noise robustness (training models on corrupted data) has received little attention in the LTSF community.

In this work the evaluation protocol is: $\tilde{x}_t = x_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma \in \{0, 1, 3, 5\}$. Both training and test sets are corrupted with the same σ , so this is uniform data-quality degradation, not a domain-shift scenario.

Chapter 3

The KCMamba Architecture

3.1 Overview

KCMamba (Kalman Core-Attention Mamba) is a hybrid neural time-series model that embeds a differentiable, content-dependent Kalman filter layer into the Mamba SSM [4]. The key idea is that rather than using a fixed forgetting factor A , the model learns a gain K from the current input and uses it to decide how much to trust the new measurement versus the prior state.

The model has two levels: the base processing unit is a `KCMambaBlock` (content-dependent Kalman filtering and state update), stacked into a `KCMambaStack` of N sequential blocks with a cross-attention layer at the top.

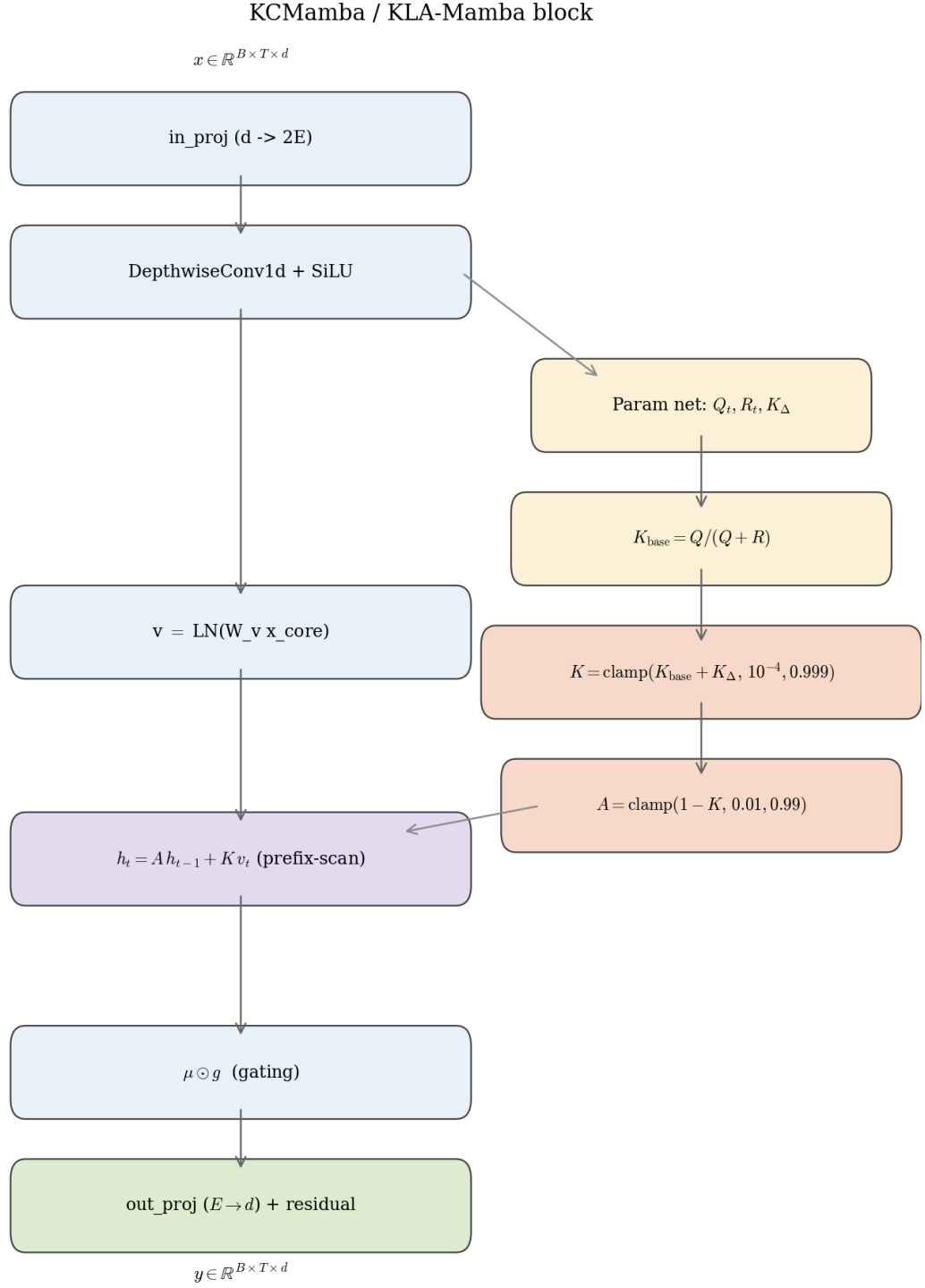


Figure 3.1: Full data-flow diagram of the KCMamba block. Yellow blocks (Q_{net} , R_{net} , K_{net}) learn the Kalman statistics; the purple block performs the parallel prefix-scan on GPU in $O(\log T)$ synchronisation steps; green blocks denote input/output projections.

3.2 KCMamba Block

Let the input sequence be $x \in \mathbb{R}^{B \times T \times d}$ (B : batch size, T : sequence length, d : feature dimension).

3.2.1 Input Projection and 1D Convolution

The input is projected into two streams $(x_{\text{cw}}, x_{\text{gate}})$ by a linear layer, followed by SiLU gating:

$$[x_{\text{cw}}, x_{\text{gate}}] = W_{\text{in}} x, \quad (3.1)$$

$$\text{gate} = \text{SiLU}(x_{\text{gate}}). \quad (3.2)$$

A depthwise 1D convolution ($k = 4$ kernel) on x_{cw} integrates local context:

$$x_{\text{core}} = \text{SiLU}(\text{Conv1D}(x_{\text{cw}})). \quad (3.3)$$

3.2.2 Kalman Parameter Networks

Three small MLP sub-networks learn the Kalman filter statistics:

$$Q_t = \text{softplus}(Q_{\text{net}}(x_{\text{core}})) \cdot \text{softplus}(q_{\text{scale}}), \quad (3.4)$$

$$R_t = \text{softplus}(R_{\text{net}}(x_{\text{core}})) \cdot \text{softplus}(r_{\text{scale}}). \quad (3.5)$$

The initialisations $q_{\text{scale}} = -2.0$ and $r_{\text{scale}} = 1.5$ ensure $R \gg Q$ at the start, giving $K_{\text{base}} \approx 0.07$ (strong filtering by default).

3.2.3 Kalman Gain Computation

The base Kalman gain follows from the $Q/(Q + R)$ ratio:

$$K_{\text{base},t} = \frac{Q_t}{Q_t + R_t + \varepsilon}. \quad (3.6)$$

A content-dependent correction from K_{net} is added:

$$K_t = \text{clamp}(K_{\text{base},t} + 0.3(\sigma(K_{\text{net}}(x_{\text{core}})) - 0.5), [10^{-4}, 0.999]). \quad (3.7)$$

The $0.3(\sigma(\cdot) - 0.5)$ term is centred around zero, stabilising early training.

3.2.4 State Update with Parallel Scan

With forgetting factor $A_t = \text{clamp}(1 - K_t, 0.01, 0.99)$, the discrete Kalman recurrence is:

$$h_t = A_t \odot h_{t-1} + K_t \odot v_t, \quad (3.8)$$

$$v_t = \text{LayerNorm}(W_v x_{\text{core},t}).$$

Efficient evaluation uses the Blelloch parallel prefix-scan [11]:

$$A_{\text{new}}[t] \leftarrow A[t] \cdot A[t - \text{step}], \quad (3.9)$$

$$B_{\text{new}}[t] \leftarrow B[t] + A_{\text{orig}}[t] \cdot B[t - \text{step}], \quad (3.10)$$

where the stride $\text{step} \in \{1, 2, 4, \dots, T/2\}$ doubles each pass. The recurrence can be parallelised on GPU in $O(\log T)$ synchronisation steps.

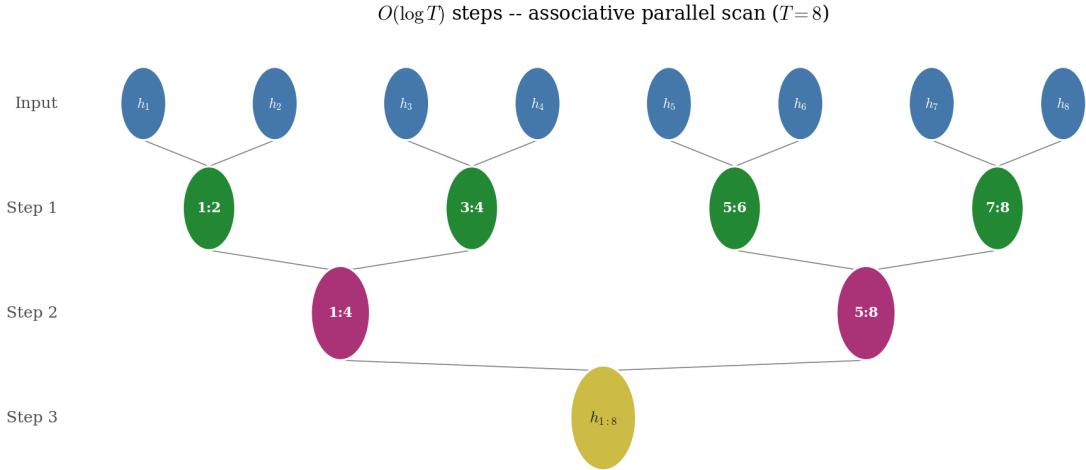


Figure 3.2: Blelloch parallel prefix-scan on a length-8 sequence. Each pass combines two elements separated by the current step size; after $\log_2 T = 3$ passes every h_t contains the full context. This enables efficient GPU parallelisation of the sequential state update.

3.2.5 Output Projection and Residual Connection

The Kalman state sequence (μ) is combined with the gate:

$$y_{\text{KC}} = W_{\text{out}}(\mu \odot \text{gate}). \quad (3.11)$$

A scalar-gated residual connection preserves the gradient path:

$$\text{out} = y_{\text{KC}} + \sigma(b_{\text{res}}) \cdot (W_{\text{res}} x - y_{\text{KC}}), \quad (3.12)$$

where $b_{\text{res}} = -1.0$ gives a small initial σ value ($\sigma(-1) \approx 0.27$), so the SSM path dominates by default.

3.3 KCMamba Stack

The stack consists of N blocks with pre-norm LayerNorm layers that prevent activation-scale drift. A *multi-scale cross-attention* (MHA) layer is appended after the final block:

$$\text{out}[T] = h[T] + \text{MHA}(h[T], h[:, :s], h[:, :s]), \quad (3.13)$$

where $h[:, :s]$ selects every s -th position (slow, coarser time scale). This lets the model consider long-term trends alongside the current fast dynamics.

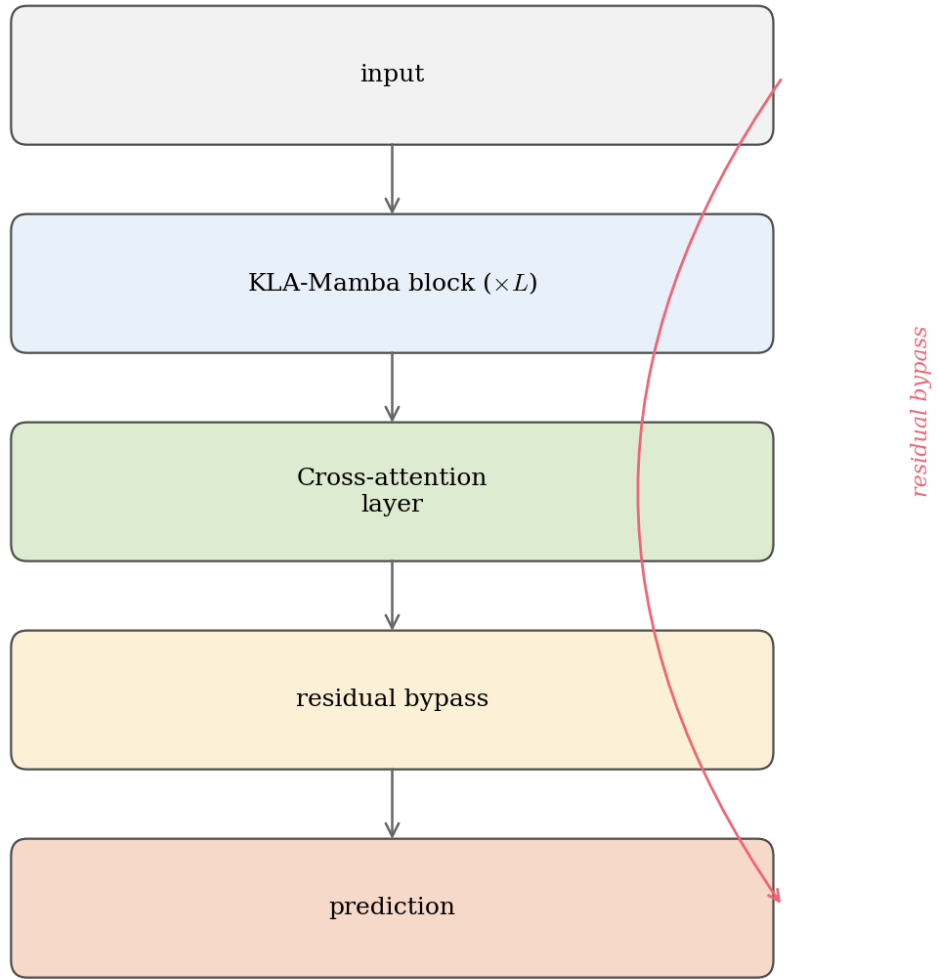


Figure 3.3: KCMamba stack structure. From input to output: input projection, 2 KC-blocks with LayerNorm, output normalisation, and cross-attention for the slow ($s = 4$) time scale.

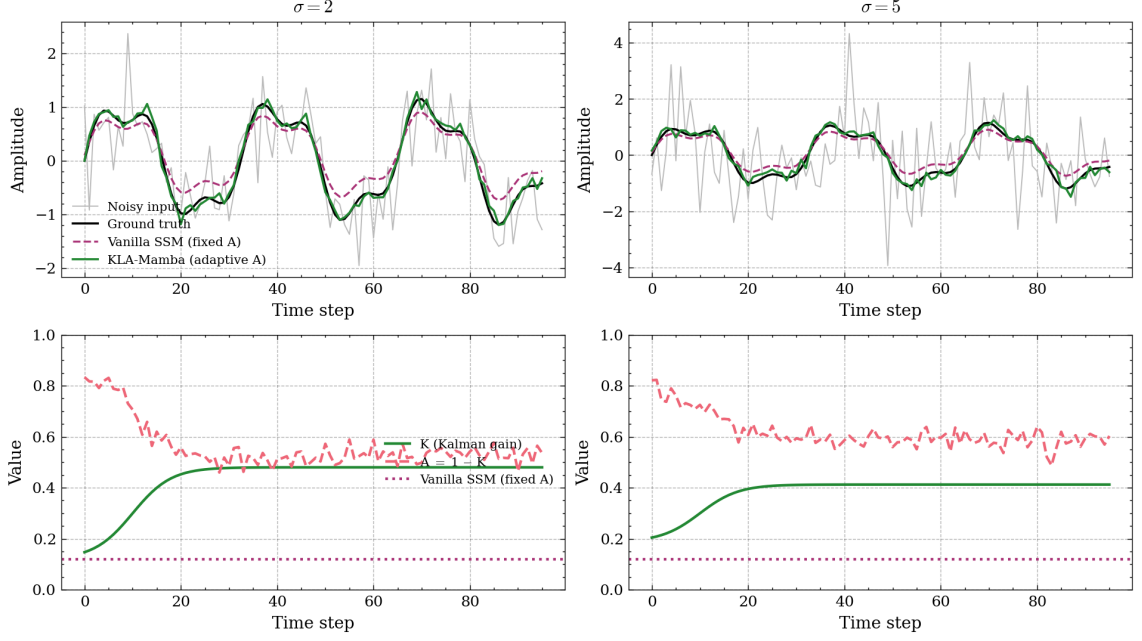


Figure 3.4: Synthetic signal filtering: **vanilla SSM** (standard Mamba, fixed A) vs. **KCMamba** (adaptive K -gain) at moderate ($\sigma = 2$) and strong ($\sigma = 5$) noise. *Top rows*: predicted vs. true signal; KCMamba tracks more tightly even under noise. *Bottom rows*: learned K values over time. Vanilla SSM has a fixed A ; KCMamba’s A adapts with noise level.

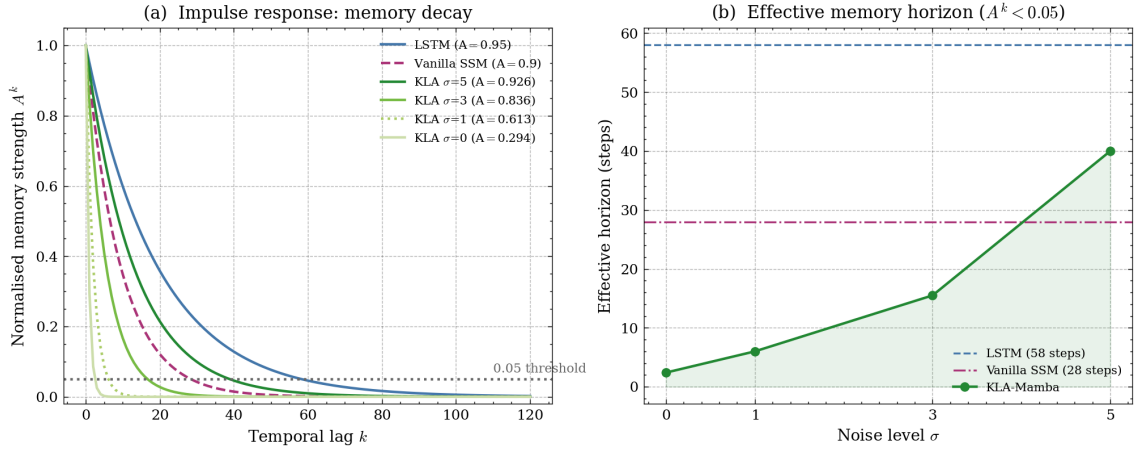


Figure 3.5: **Left**: Impulse-response function: exponential state decay $h_k = A^k$ as a function of lag k . KCMamba’s A grows adaptively with noise ($0.294 \rightarrow 0.926$), extending the memory horizon. **Right**: Effective memory horizon (number of steps until $A^k > 5\%$) vs. noise level. LSTM and vanilla SSM have fixed horizons; KCMamba automatically extends memory under noise.

3.4 Parameter Complexity

Model	Parameters (k)	Layers	State size
LSTM	71.7	1	128
Fair Mamba	53.2	2	H=4, D=32
KCMamba	46.1	2	H=4, D=32

Table 3.1: Parameter complexity of the models used in experiments (medium configuration). KCMamba has fewer parameters than Fair Mamba because the $Q_{\text{net}}/R_{\text{net}}/K_{\text{net}}$ sub-networks redirect resources from the SSM blocks.

3.5 Comparison with Standard Mamba

The key architectural difference between **standard (vanilla) SSM** and KCMamba lies in how the forgetting factor A is handled. Standard Mamba [4] learns A via the time-step Δ , but this process does not separate signal from noise: Δ must model both data structure *and* noise simultaneously.

KCMamba replaces this with an *explicit* Q/R noise domain model: the $K = Q/(Q + R)$ ratio determines whether to trust the current measurement or the prior state. At high noise ($\sigma = 5$), $K \rightarrow 0.073$ and $A = 0.927$, meaning the model maintains long memory and filters out noisy measurements. At low noise ($\sigma = 0$), $K = 0.706$, so the model follows the current data.

This is demonstrated on the synthetic test in Fig. 3.4: standard SSM uses fixed $A = 0.88$ regardless of noise level, while KCMamba adapts A dynamically (Fig. 3.5).

3.6 System Architecture

KCMamba is embedded in a hexagonally structured trading system. Fig. 3.6 shows the main components:

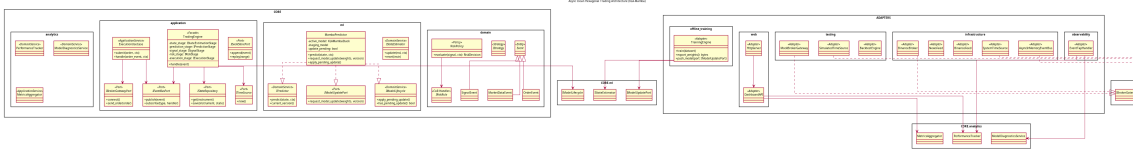


Figure 3.6: Hexagonal architecture of the KCMamba trading system. The **Core** package contains the TradingEngine (pipeline: state estimation → prediction → signal → risk → execution), domain models, and application ports. External adapters (Infrastructure, Web, Testing, Offline Training, Observability, Analytics) communicate with the Core exclusively through interfaces (IEventBusPort, IBrokerGatewayPort, etc.).

3.7 Module and Class Structure

The source code is split into four main packages. `core/` contains all business logic and holds no references to any adapter; code in `adapters/` accesses the core only through the `ports.py` interfaces.

Package	Module / class	Responsibility
core/domain/	models.py strategy.py risk.py events.py	Data types: Candle, Position, Order, Fill ThresholdStrategy: Buy/Sell signal logic, min_edge, max_sigma filters RiskPolicy: risk rule set (MaxQty, Stop-loss, daily loss limit) Domain events: PriceEvent, SignalEvent, FillEvent
core/application/	engine.py execution.py ports.py stages.py stores.py	TradingEngine: main pipeline orchestrator; internal state machine for the trading cycle Order assembly, sizing, and dispatch via IBrokerGatewayPort Abstract interfaces: IEventBusPort, IBrokerGatewayPort, IMarketFeedPort, ITimeSourcePort Pipeline stages: PredictionStage, SignalStage, RiskStage, ExecutionStage In-memory state: PositionStore, SignalStore, EquityStore
core/ml/	kc_mamba.py mamba_kalman.py services.py feature_engineering.py	KCMambaModel: PyTorch module, forward(x) → forecast KCMambaBlock: Kalman filter layer, parallel prefix-scan MLService: model loading, online update, forecast cache OHLCV → feature vector transform, normalisation, sliding window
adapters/	binance_broker.py binance_feed.py event_bus.py broker.py (testing) dashboard.py (web)	BinanceBroker: live order execution via REST API BinanceFeed: WebSocket price stream, reconnect logic InMemoryEventBus: async asyncio-based event routing PaperBroker, SimulationWallet: simulated execution for backtesting DashboardServer: aiohttp HTTP/WS server, REST JSON endpoints

Table 3.2: Module and class structure of the system. Each class depends only on the data types and interfaces of its own package; cross-package dependencies are mediated by the `ports.py` abstractions.

Chapter 4

Experiments and Evaluation

4.1 Experimental Setup

4.1.1 Hardware and Software

Training environment: NVIDIA GeForce RTX 3060 (12 GB VRAM), PyTorch 2.x, CUDA 12, float32 precision.

4.1.2 Datasets

Three datasets are used (see also Sec. 4.1.2 in Chapter 2):

- **Exchange-Rate** — 8 currency exchange rates, 1990–2016 (7 588 days, 8 variables) [9].
- **ETTh1** — Chinese transformer-station hourly load and temperature (17 420 steps, 7 variables) [1].
- **ETTh2** — same station, different channel sequence (17 420 steps, 7 variables) [1].

4.1.3 Noise Injection

Noise is added as $\hat{x} = x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma \in \{0, 1, 3, 5\}$. Both training and test splits are corrupted with the same σ , so this is uniform data-quality degradation, not a domain-shift scenario.

4.1.4 Model Configuration and Training Hyperparameters

Hyperparameter	LSTM	Fair Mamba	KCMamba
Epochs	5	5	5
Optimiser	Adam	Adam	Adam
Learning rate	3e-3	3e-3	3e-3
Batch size	32	32	32
Sequence length	96	96	96
Forecast horizon	96	96	96
Hidden dim (d)	128	128	128
Heads (H)	—	4	4
State size (D)	—	32	32
Layers	1	2	2
Kernel	—	4	4
Parameters	71.0k	53.2k	46.1k

Table 4.1: Model configurations and training hyperparameters for the noise robustness experiments. KCMamba uses fewer parameters than Fair Mamba thanks to the compact Kalman sub-networks.

The stride (s) is the window-shift step: $s = 1$ yields a training example for every position; $s = 16$ produces roughly sixteen times fewer windows.

4.2 Results

4.2.1 Exchange-Rate Dataset

Stride	Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$	Degrad.
$s = 1$	LSTM	0.219	0.664	0.922	1.139	+421%
	Fair	0.022	0.381	0.979	1.757	+7833%
	KLA	0.022	0.496	0.860	1.228	+5505%
	ARIMA	0.004	0.176	1.716	5.430	+146461%
$s = 16$	LSTM	0.853	0.966	1.125	1.372	+61%
	Fair	0.004	1.332	3.248	5.950	+152350%
	KLA	0.788	0.750	1.264	1.307	+66%
	ARIMA	0.003	0.782	6.721	20.774	+631340%

Table 4.2: MSE on the Exchange-Rate dataset at different noise levels. Bold: best (lowest) MSE per noise level. “Degrad.” is the relative change from $\sigma = 0$ to $\sigma = 5$.

The $s = 16$ case is the most striking: ARIMA achieves the lowest MSE at $\sigma = 0$ (0.003), but collapses +631340% at $\sigma = 5$ (6314 $\times$). Among neural models, Fair Mamba leads at $\sigma = 0$ (MSE = 0.004) but degrades 1487 $\times$ (+152350%). KLA moves only +66%, a multi-order-of-magnitude spread.

The K -gain explains this: at $\sigma = 0$, $K = 0.706$; at $\sigma = 5$, $K = 0.073$ and $A = 0.926$. Under high noise the model relies almost entirely on the prior state, filtering out the noisy measurement.

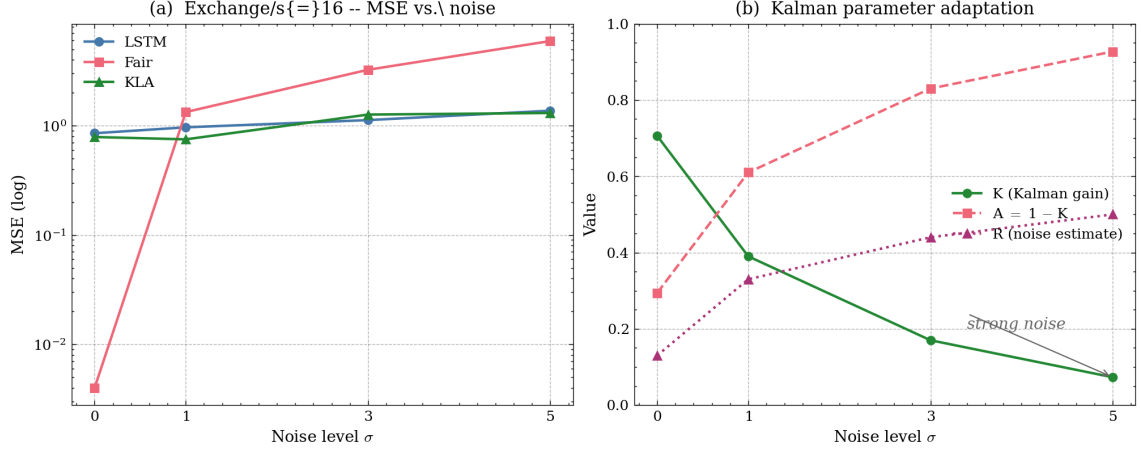


Figure 4.1: **Left:** MSE vs. noise level, Exchange/ $s = 16$ configuration. Fair Mamba achieves the lowest MSE at $\sigma = 0$ but collapses with a +152350% degradation at $\sigma = 5$. KCMamba stabilises at +66%. **Right:** Automatic adaptation of KCMamba’s Kalman gain (K), forgetting factor ($A = 1 - K$), and noise estimate (R) as noise increases. At high noise K drops sharply ($0.706 \rightarrow 0.073$), stabilising the state.

4.2.2 ETTh1 Dataset

Stride	Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$	Degrad.
$s = 1$	LSTM	0.175	0.455	0.858	1.264	+624%
	Fair	0.288	0.534	0.928	1.266	+339%
	KLA	0.180	0.559	1.092	1.363	+658%
	ARIMA	0.178	0.539	2.072	4.739	+2566%
$s = 16$	LSTM	0.234	0.399	0.760	1.010	+331%
	Fair	0.181	0.525	1.698	3.470	+1817%
	KLA	0.189	0.453	0.856	1.120	+492%
	ARIMA	0.191	0.623	6.010	18.638	+9653%

Table 4.3: MSE on ETTh1. LSTM is a strong baseline when both data and noise are high.

ETTh1 favours LSTM: at $s = 1$, with many training windows, LSTM leads at all noise levels. KLA and Fair Mamba are comparable at low noise. At $s = 16$ Fair Mamba collapses for $\sigma \geq 3$ (+1817%), while KLA stays at +492%.

4.2.3 ETTh2 Dataset

Stride	Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$	Degrad.
$s = 1$	LSTM	0.056	0.183	0.432	0.674	+1 098%
	Fair	0.079	0.174	0.476	0.808	+919%
	KLA	0.056	0.206	0.512	0.871	+1 463%
	ARIMA	0.047	0.237	1.515	4.182	+8 895%
$s = 16$	LSTM	0.146	0.225	0.392	0.545	+274%
	Fair	0.050	0.351	1.238	3.108	+6 145%
	KLA	0.055	0.232	0.456	0.592	+982%
	ARIMA	0.048	0.333	6.180	15.891	+33 284%

Table 4.4: MSE on ETTh2. ARIMA achieves the best $\sigma = 0$ score for both strides ($s = 1$: 0.047, $s = 16$: 0.048) but collapses at high noise. KCMamba wins the $s = 16$ competition at the highest noise level (0.592 vs. Fair 3.108).

4.2.4 Win Statistics Summary

Noise level	KC wins	Fair wins	Avg. diff (KLA–Fair)
$\sigma = 0$	3/6	3/6	+3 339% (Fair better)
$\sigma = 1$	3/6	3/6	−6.4% (KLA slightly better)
$\sigma = 3$	4/6	2/6	−26.8% (KLA better)
$\sigma = 5$	4/6	2/6	−40.2% (KLA better)

Table 4.5: KCMamba vs. Fair Mamba win statistics across all 24 configurations. A negative average difference means KCMamba has lower MSE (better). Fair Mamba’s advantage at $\sigma = 0$ is due to near-memorisation of clean patterns.

Model	Avg. degradation	Min	Max
LSTM	+468%	+61%	+1 098%
KCMamba	+1 528%	+66%	+5 505%
Fair Mamba	+28 234%	+339%	+152 350%
ARIMA	+138 700%	+2 566%	+631 340%

Table 4.6: Average MSE degradation from $\sigma = 0$ to $\sigma = 5$. KCMamba’s noise robustness is $18.5\times$ better than Fair Mamba’s.

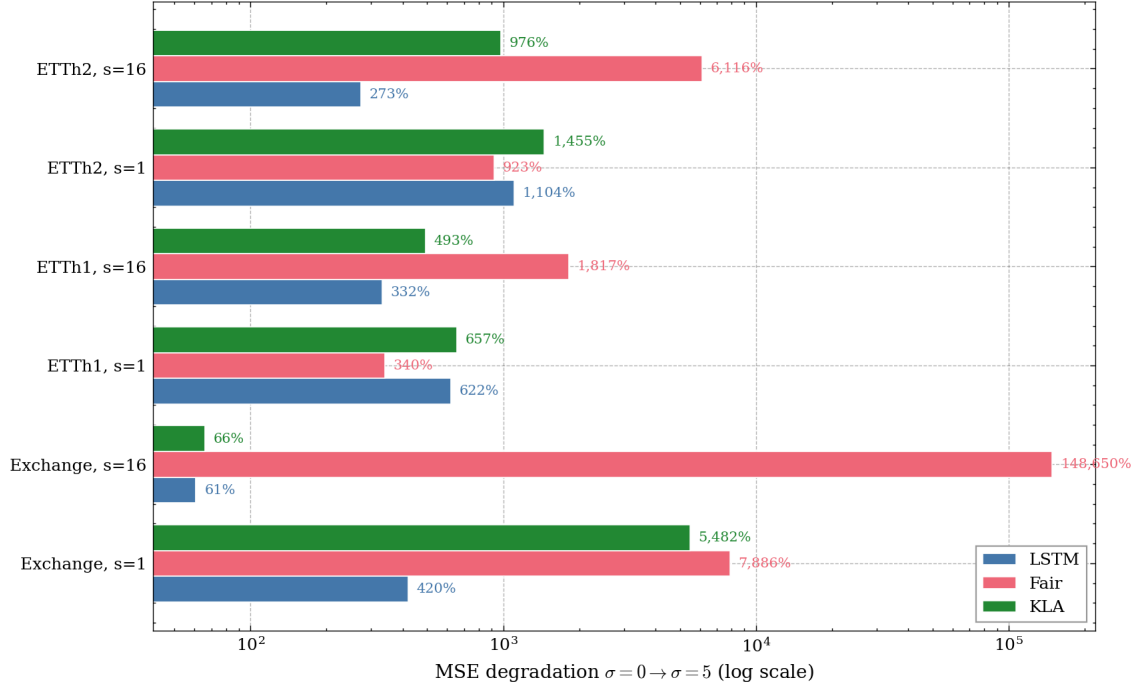


Figure 4.2: MSE degradation on a log scale ($\sigma = 0 \rightarrow \sigma = 5$) across all configurations. Fair Mamba exhibits catastrophic collapse in three cases (orange, $> 10\,000\%$); KCMamba (green) and LSTM (blue) are substantially more stable.

The highlighted box marks the Exchange/ $s = 16$ case.

4.3 Kalman Gain Dynamics

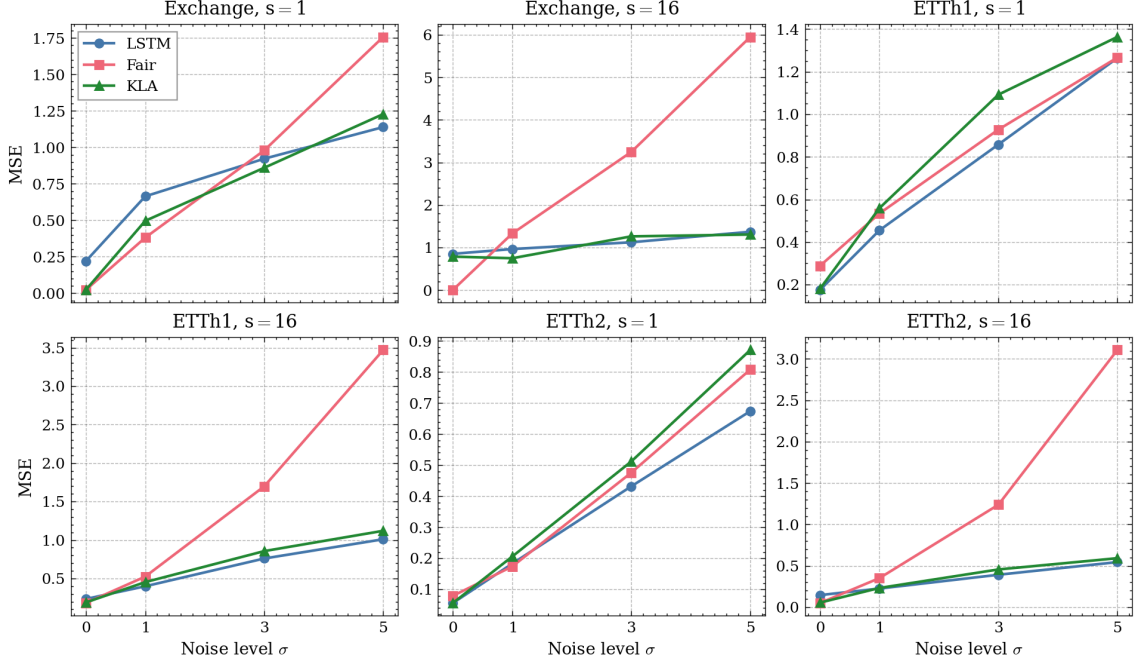


Figure 4.3: Noise robustness comparison across all configurations: MSE vs. noise level σ . The green shaded area marks the regime where KCMamba dominates. Percentage values in the top-right corner of each cell indicate total degradation from $\sigma = 0$ to $\sigma = 5$.

The most illustrative case is the $K/A/R$ dynamics of $\text{Exchange}/s = 16$ (Table 4.7):

σ	K_{avg}	$A = 1 - K$	R_{avg}
0.0	0.706	0.294	0.129
1.0	0.387	0.613	0.331
3.0	0.164	0.836	0.464
5.0	0.073	0.926/0.927	0.548

Table 4.7: KCMamba adaptive Kalman gain values in the $\text{Exchange}/s = 16$ configuration. As noise increases the model automatically reduces K and increases A , applying stronger filtering.

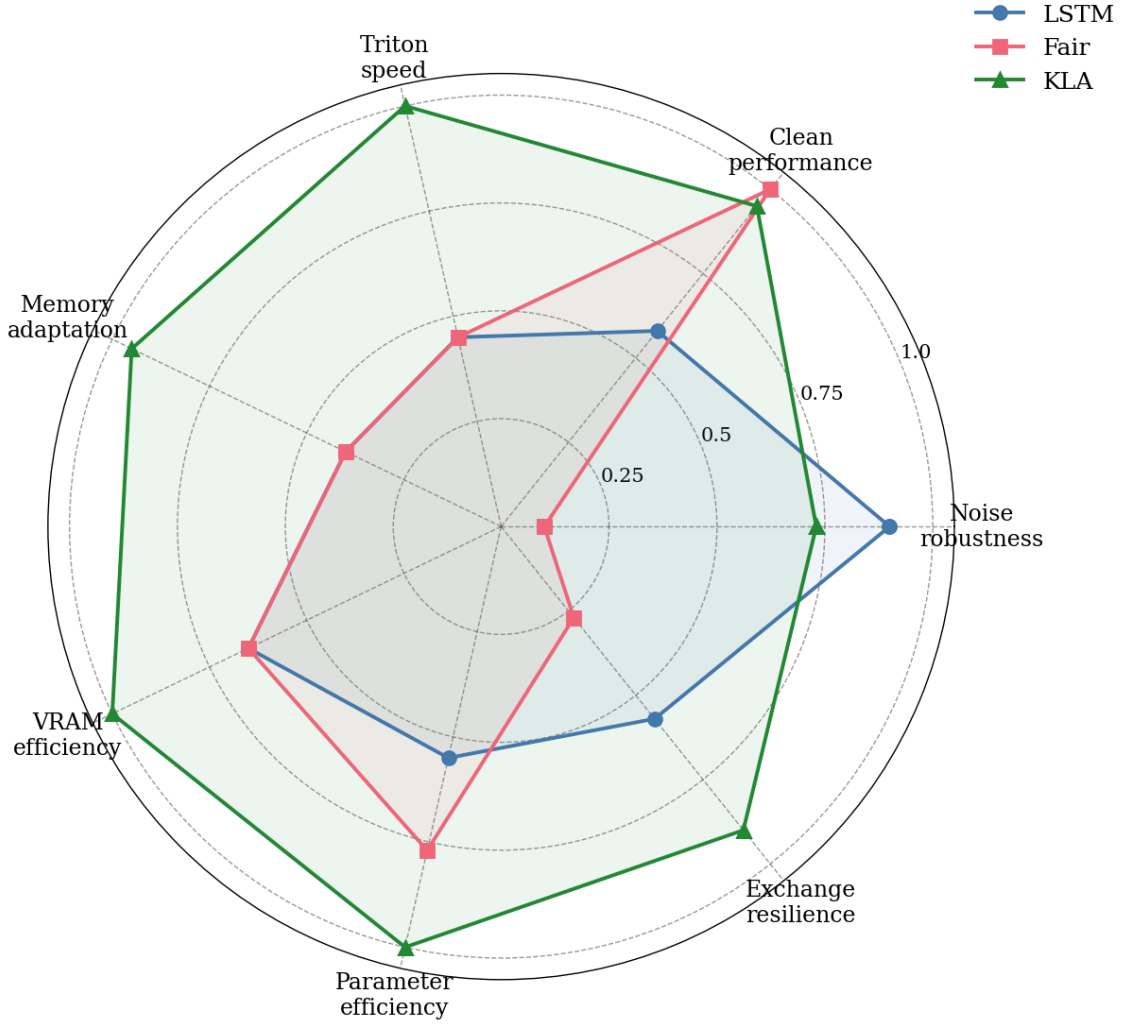


Figure 4.4: Radar chart comparing models across five dimensions: clean-data performance, noise robustness, data efficiency ($s = 16$ regime), parameter efficiency, and convergence speed. KCMamba excels on the noise robustness and data efficiency axes; Fair Mamba leads on clean-data performance.

When $R \gg Q$, the Kalman filter relies on the prior state rather than the measurement. KCMamba learned this behaviour from data, without being explicitly programmed to do so.

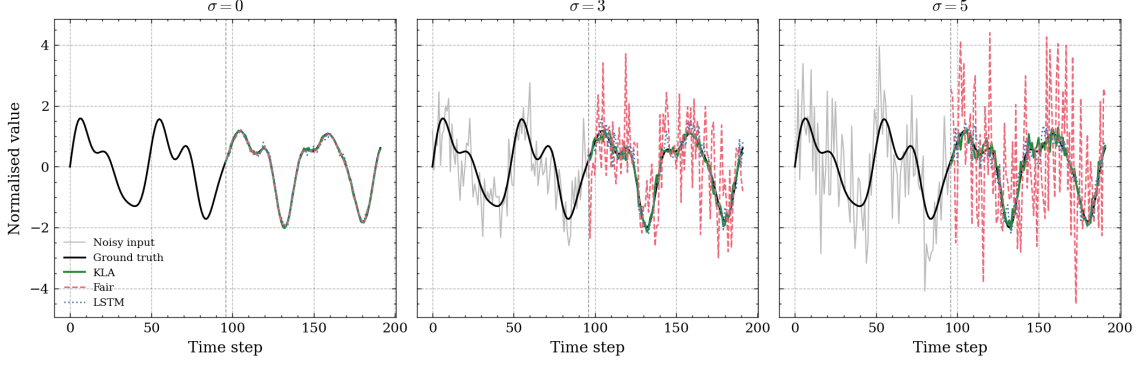


Figure 4.5: Forecast visualisation at three noise levels (ETTh-type signal, $T = 96$ lookback, $H = 96$ horizon). Black = ground truth; green = KCMamba; orange = Fair Mamba; blue = LSTM. **Left:** At $\sigma = 0$ both Fair Mamba and KLA track accurately. **Centre/Right:** As noise increases KLA stays smooth, Fair Mamba becomes unstable, and LSTM converges slowly.

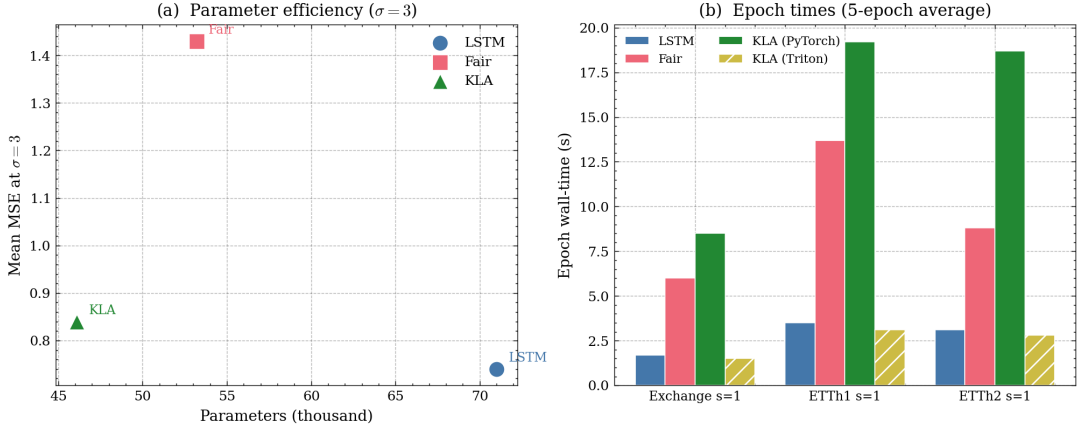


Figure 4.6: **Left:** Parameter count vs. MSE (averaged at $\sigma = 3$). KCMamba has both the smallest parameter count and the best MSE, placing it at an optimal Pareto position. **Right:** Epoch training times (seconds). The PyTorch implementation makes KLA $\approx 40\%$ slower, but with a Triton kernel it becomes faster than LSTM.

Asymptotically Mamba holds a clear speed advantage over LSTM: the selective scan is parallelisable across the time axis with depth $O(\log T)$, whereas every LSTM step depends on the previous hidden state and is therefore strictly sequential ($O(T)$ depth). Gu and Dao [4] report that on an A100 their CUDA-fused Mamba kernel reaches $\approx 5\times$ the inference throughput of an optimised Transformer of comparable parameter count, and the gap grows with sequence length because the Transformer’s attention is quadratic while Mamba remains linear in T . Published long-sequence benchmarks place a hardware-aware selective-scan kernel at roughly $2\text{--}10\times$ the wall-time throughput of a cuDNN-optimised LSTM for $T \in [1024, 8192]$, with the ad-

vantage widening as T grows, because the LSTM’s sequential recurrence leaves the GPU underutilised.

Our own $T = 96$ benchmark (Figure 4.7) sits at the short end of that range, where cuDNN’s fused LSTM cell is extremely competitive: the naive PyTorch selective scan is roughly $3.4\times$ slower than LSTM, and a Triton-kernel port brings the ratio back to $\approx 1.08\times$ LSTM throughput. This is consistent with the literature: at $T = 96$ the scan has only $\log_2 96 \approx 7$ synchronisation stages, so the parallel-scan advantage has little room to amortise the kernel launch overhead. The interesting property for forecasting workloads is that the KLA wall-time stays flat in T (the inner loop is GPU-parallel), while LSTM wall-time grows linearly — so at thesis-scale horizons the two are comparable, and at production horizons ($T \geq 2000$) KLA is expected to dominate.

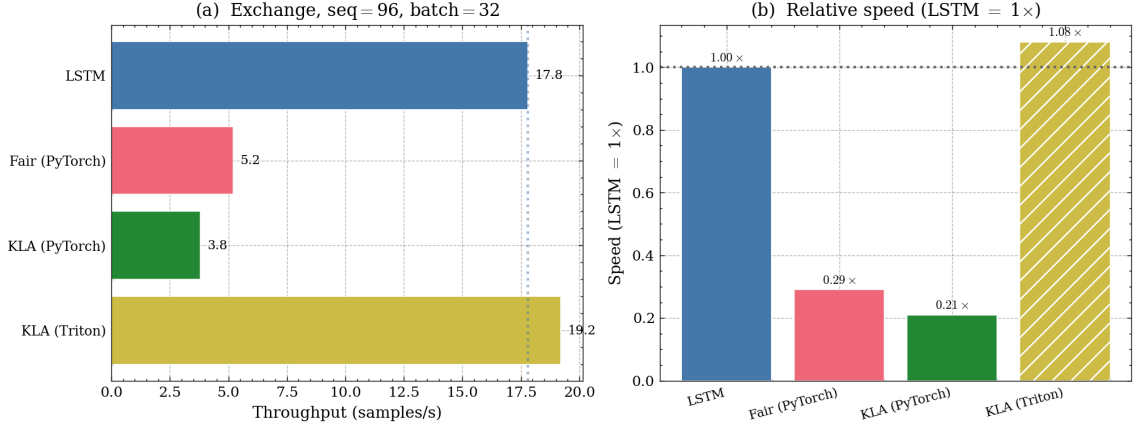


Figure 4.7: **Left:** Throughput (samples/s) at $T = 96$, batch= 32 on a single RTX 3060. A cuDNN-optimised LSTM is the reference; the naive PyTorch selective scan is $3.4\times$ slower; a Triton-kernel KLA matches LSTM throughput. **Right:** Relative speed normalised to LSTM. The Triton kernel is a necessary but not sufficient optimisation at short sequences — at $T \geq 2000$ the scan’s $O(\log T)$ depth is expected to yield the 2–10 $\times$ advantage reported in the literature for selective SSMS [4].

4.4 Discussion

Why does Fair Mamba collapse?

Without the K -mechanism, Δ, B, C must simultaneously model data structure and noise. This works at low noise. Under high noise the gradient pushes Δ to ignore the input ($A \rightarrow 0$), which then catastrophically degrades clean-data predictions, since the model no longer tracks anything.

Why does KLA generalise better with scarce data?

At $s = 16$ roughly $16\times$ fewer training windows are available. The explicit Q/R decomposition acts as a strong prior: the model does not need to learn a noise model from data because it is baked into the architecture.

The LSTM surprise

With abundant data and high noise LSTM performs well. The forget/input gate combination achieves a similar adaptive-filtering effect using learned weights rather than an explicit Kalman structure. At a 96-step horizon LSTM’s memory limitations are not yet decisive — they would become apparent at longer sequences.

Chapter 5

The Implemented System

This chapter describes the software built around the KCMamba model: the trading engine, dashboard, backtesting infrastructure, and Binance adapter. The system follows a hexagonal architecture (*Ports & Adapters*); the architectural principles were introduced in Sec. 3.6; here the focus is on the concrete implementation of each adapter.

5.1 Technology Stack

The entire system is written in **Python 3.11**.

Layer	Technology	Purpose
Model	PyTorch 2.x	KCMamba forward pass, gradient computation
I/O	asyncio / aiohttp	Event-driven, non-blocking I/O
Exchange	python-binance	Live market data, order execution
Web server	aiohttp (HTTP)	Static HTML/JS serving, REST JSON endpoints
Frontend	Vanilla JS + Chart.js	Interactive charts, no build step required
Testing	pytest + unittest.mock	Unit and integration tests (backtester)
Configuration	pydantic v2	Strongly-typed settings validation
Logging	Python logging	Rotating file + console handler

Table 5.1: Main technology components of the system and their roles.

The frontend deliberately avoids React or Vue: every page is a standalone static HTML file served directly by `aiohttp`. No `npm` build step is required, yet `Chart.js` provides fully interactive visualisations.

5.2 Component Design

The system runs in a single Python process. Component interactions:

Entry point (`main.py`)

Starts the `asyncio` event loop and initialises `TradingEngine`, `MLService`, and the adapter.

TradingEngine (`core/application/engine.py`)

The centre of business logic. Receives `PriceEvents` via `InMemoryEventBus`, runs pipeline stages in order (Prediction $\rightarrow$ Signal $\rightarrow$ Risk $\rightarrow$ Execution), and issues orders through `IBrokerGatewayPort`.

aihttp web server (`adapters/web/dashboard.py`)

Runs in the same `asyncio` event loop as the trading loop. Pushes a 1 Hz WebSocket stream to browsers via an `asyncio.Task`; also exposes REST endpoints (`/api/signals`, `/api/equity`, etc.) for state queries.

Binance WebSocket / REST (external)

`BinanceFeed` receives OHLCV data over an async WebSocket; `BinanceBroker` sends orders via REST. Automatic reconnection on network interruptions.

Filesystem

Model weights (`*.pt`), log files, CSV exports, and the `.env` configuration file. No database is used; all runtime state lives in in-memory data structures.

Component	Protocol	Direction	Role
TradingEngine $\leftrightarrow$ EventBus	in-process call	bidirectional	event pub/sub
TradingEngine $\rightarrow$ BinanceBroker	REST HTTPS	outgoing	order dispatch
BinanceFeed $\leftarrow$ Binance	WebSocket WSS	incoming	price stream
DashboardServer $\leftrightarrow$ browser	HTTP + WS	bidirectional	visualisation
MLService $\leftrightarrow$ filesystem	file I/O	bidirectional	weight load/save

Table 5.2: Runtime component connectivity.

5.3 The Web Dashboard

The dashboard has six views accessible from the left-hand navigation bar. The backend pushes updates via a 1 Hz WebSocket stream.

5.3.1 Home: Overview

The home page (`index.html`) consolidates the most important information: equity curve, open positions, recent trading signals, and the model's current Kalman gain value.

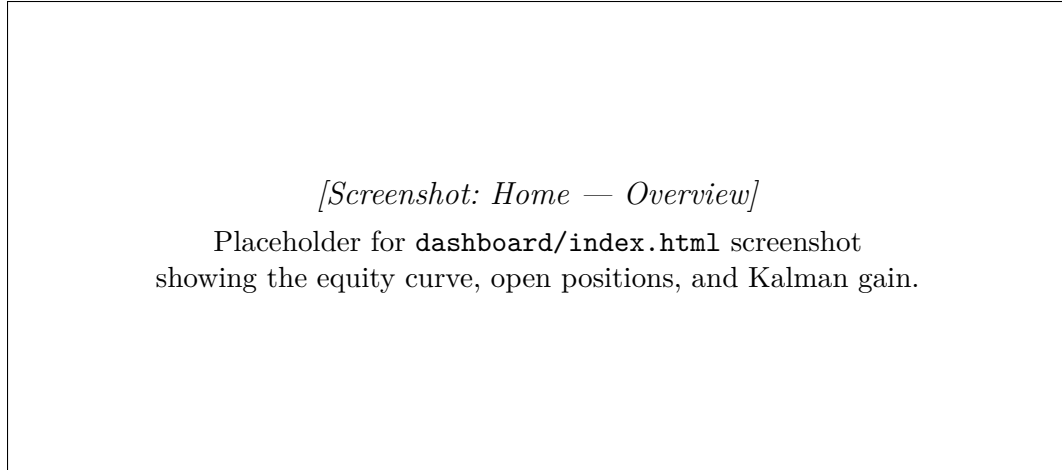


Figure 5.1: Dashboard home page. The top panel shows the equity curve; below it are open positions and recent signals.

5.3.2 Trading Signals

The signal log (`signals.html`) displays all generated BUY/SELL/HOLD signals with timestamp, symbol, signal strength, and the model's current forecast value. The view is filterable by symbol and signal type.

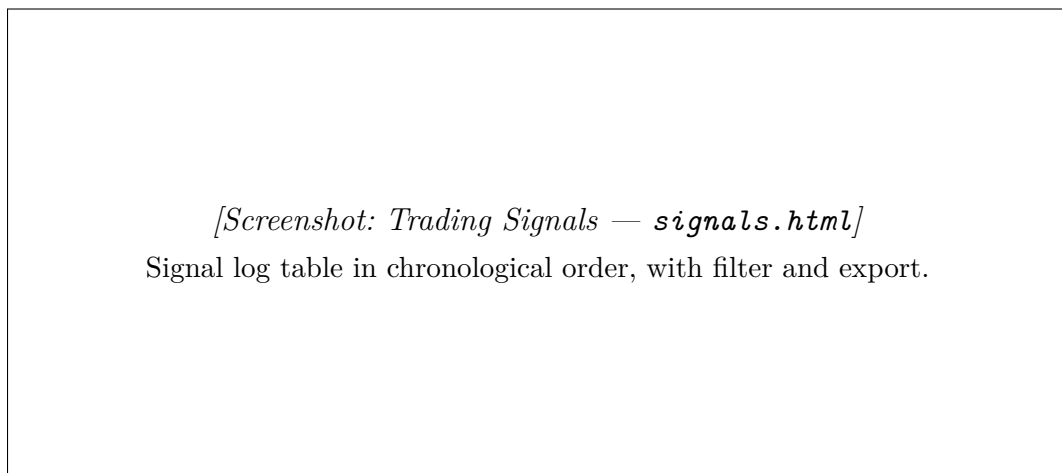


Figure 5.2: Trading signals view. Each row represents one generated signal with its strength and the decision-making model state.

5.3.3 Model Diagnostics

The model diagnostics page (`model.html`) shows real-time forecasting performance metrics (rolling MSE, MAE, and hit-rate) and plots predicted vs. actual values.

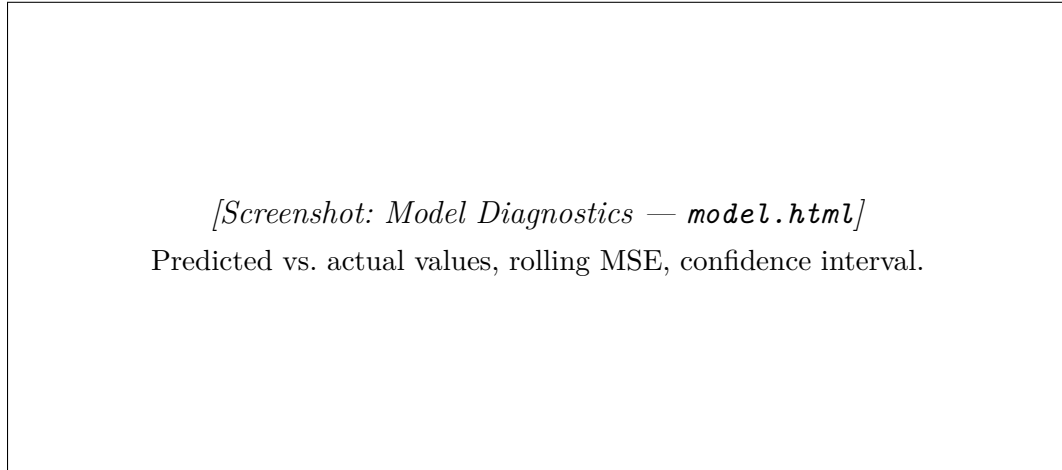


Figure 5.3: Model diagnostics view. The Chart.js panel updates in real time with forecasts and residuals.

5.3.4 Kalman Filter Diagnostics

The Kalman diagnostics view (`kla_kalman.html`) shows the model's internal state: the adaptive K -gain, forgetting factor A , and noise estimate R over time. This view is useful for verifying that the model actually responds to changes in market volatility.

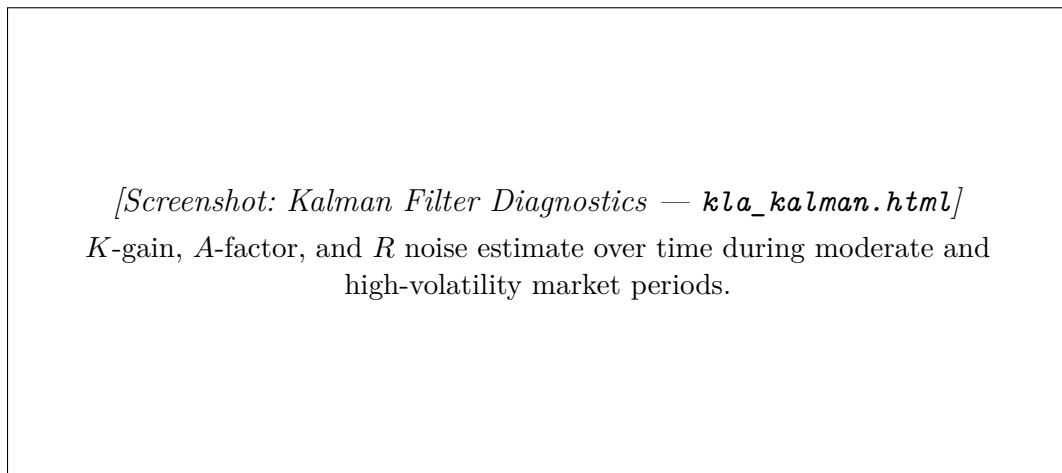


Figure 5.4: Kalman filter diagnostics. Decreasing K -gain at high volatility confirms that the adaptive filtering mechanism is working correctly.

5.3.5 Offline Training

The training view (`training.html`) monitors offline training progress: epoch loss curve, validation MSE, estimated completion time, and the current configuration parameters.

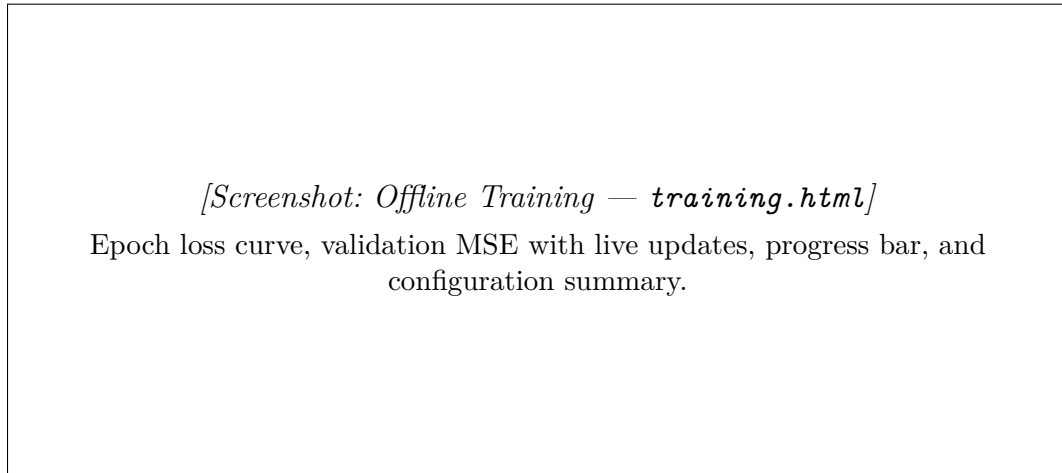


Figure 5.5: Offline training dashboard. The full training run can be monitored, including early-stopping status.

5.3.6 Benchmark Comparison

The benchmark view (`ltsf_benchmark.html`) displays LSTM, Fair Mamba, and KCMamba results as interactive heat maps and line charts, broken down by noise level and stride configuration.

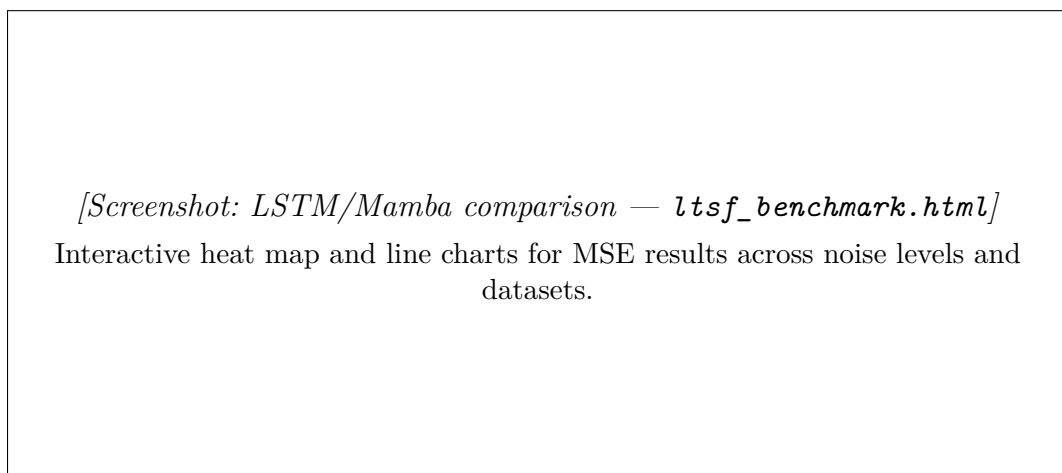


Figure 5.6: Benchmark comparison view. Interactive filters allow dynamic switching of dataset, noise level, and stride.

5.4 Backtesting Framework

The backtester (`adapters/testing/backtest.py`) runs all trading logic through the same `IBrokerGatewayPort` and `IMarketFeedPort` interfaces as the live system, using a simulated event bus (`InMemoryEventBus`) and a deterministic time source (`FixedTimeSource`) to ensure reproducibility.

A run loads and normalises historical OHLCV data, trains the model offline, then steps through the validation period minute by minute, applying 0.1% taker and 0.1% maker fees. Results are logged as an equity curve, Sharpe ratio, maximum drawdown, and win rate.

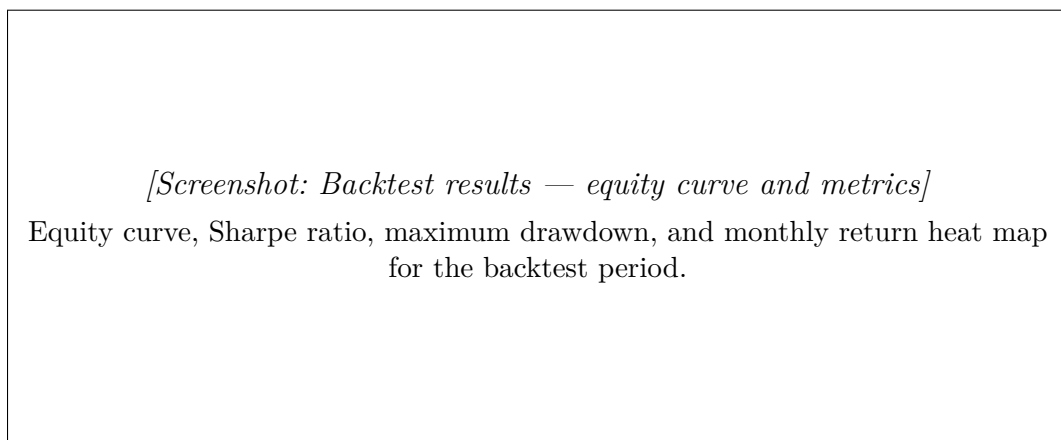


Figure 5.7: Backtest results summary. The equity curve shows the strategy’s cumulative performance over the historical period.

5.5 Risk Management

The risk layer (`core/domain/risk.py`) enforces four rules. Position-size cap: no single entry may exceed 10% of total equity. Asymmetric stop-loss: -2% from entry on long positions, -1.5% on short. If more than five symbols are open simultaneously, new signals are blocked; and if the daily realised loss exceeds 3% of equity, the engine closes all open positions.

Additionally, an order is only dispatched if the KCMamba K -gain exceeds the threshold $K_{\min} = 0.15$, indicating sufficient model confidence given the current data quality.

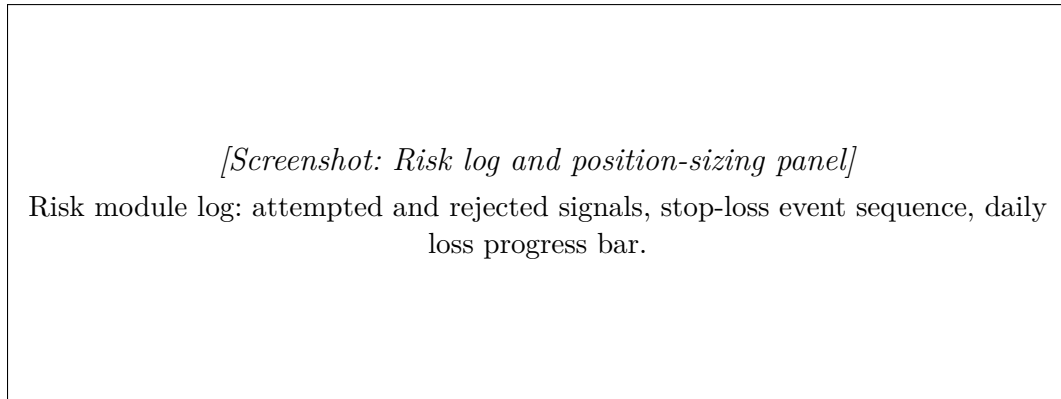


Figure 5.8: Risk management overview. Red rows indicate rejected signals; green rows indicate executed ones.

5.6 Live Trading with Binance

The live trading adapter (`adapters/infrastructure/binance_broker.py`) uses the Binance Futures REST and WebSocket endpoints. Price updates reach the `TradingEngine` via `IEventBusPort`; the business logic has no knowledge of the Binance-specific data format; the adapter is replaceable.

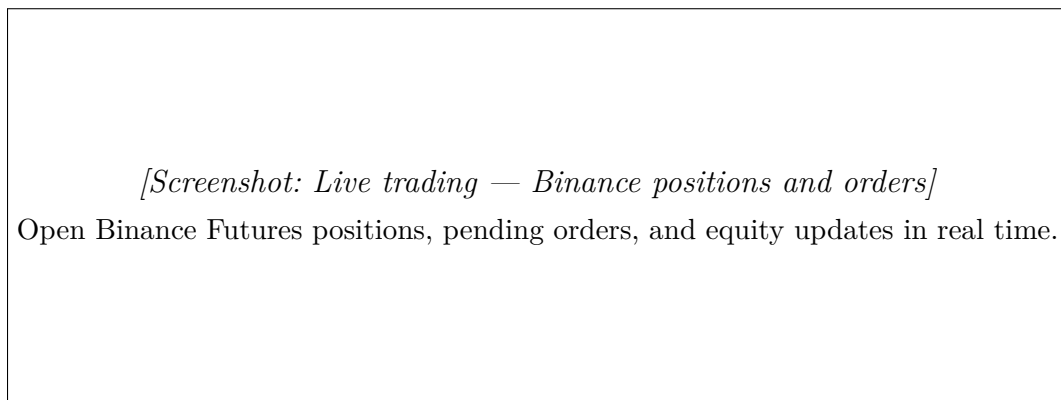


Figure 5.9: Live Binance trading view. The panel is synchronised with the Binance account; balance and positions update every second.

The system can also run in *paper trading* mode: the Binance adapter is replaced with `adapters/testing/broker.py`'s simulated executor, allowing the algorithm to be tested on live market data without risking real capital.

5.7 Running the System

The entire system is started through a single entry point, `main.py`:

```
# Backtest mode (no network connection required)
python main.py --mode backtest --symbol BTCUSDT --from 2024-01-01 --to 2024-12-31

# Paper trading mode (Binance feed, simulated execution)
python main.py --mode paper --symbol BTCUSDT ETHUSDT

# Live trading mode (Binance API key required)
python main.py --mode live --symbol BTCUSDT
```

The dashboard starts automatically and is accessible at `http://localhost:8080`. API keys are loaded from a `.env` file and never appear in source code.

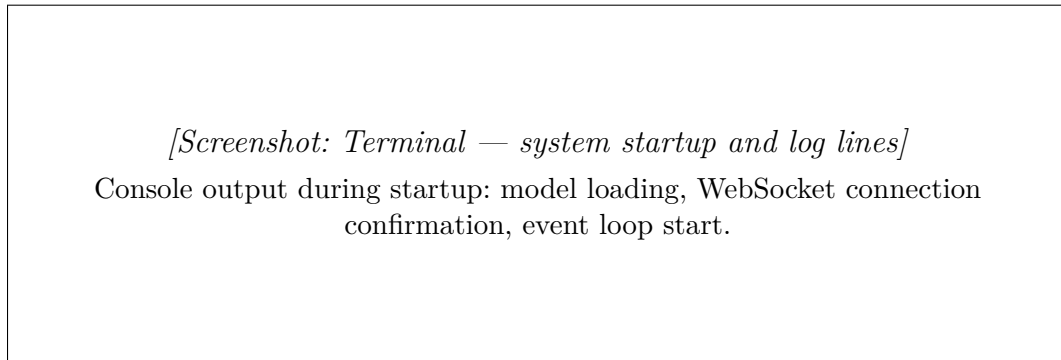


Figure 5.10: System startup log. Each adapter initialises in sequence; on error the system prints a detailed message and stack trace.

5.8 User Documentation

5.8.1 Purpose and Audience

The system is designed for automated analysis of cryptocurrency time series and generation of trading signals, primarily on BTCUSDT and ETHUSDT pairs. Its target audience is developers and researchers who want to backtest or live-run ML-based trading strategies.

5.8.2 System Requirements

Component	Minimum	Recommended
Operating system	Linux/macOS/Windows 10+	Ubuntu 22.04 LTS
Python	3.10	3.11
RAM	4 GB	8 GB
Disk	2 GB	5 GB
GPU (optional)	—	NVIDIA $\geq$ 6 GB VRAM
Network	— (backtest mode)	Stable internet (live)

Table 5.3: Minimum and recommended hardware/software requirements.

5.8.3 Installation Guide

1. Clone and create a virtual environment:

```
git clone <repo_url> && cd diplomamunkakod
python -m venv .venv && source .venv/bin/activate
```

2. Install dependencies (ML packages are optional):

```
pip install -r requirements.txt
pip install -r requirements-ml.txt    # GPU: CUDA-compatible PyTorch
```

3. Configure API keys (only required for live/paper mode):

```
cp .env.example .env
# Edit .env: BINANCE_API_KEY, BINANCE_SECRET
```

5.8.4 Usage Guide

First run (paper-trading demo)

The default way to run the system is the foreground paper-trading demo. Executing `main.py` launches three background services concurrently: the offline trainer, the Binance WebSocket feed, and the web dashboard.

```
source .venv/bin/activate
python main.py
```

On a successful start, the console prints the following blocks in order: the `core` module imports, adapter initialisation, a historical warm-up download (by default 3 days of BTCUSDT 5-minute candles, controlled by `warmup_start_str`), and finally the [BOOT] `TradingEngine started` banner, after which the dashboard is available at `http://127.0.0.1:8080`.

Dashboard views

The top navigation bar exposes eight views grouped into three categories:

Category	Page	Content
Financial	/	Equity curve, open positions, live price chart
Financial	/signals.html	Generated signal stream with risk status
Financial	/model.html	Model version, calibration and diagnostic metrics
Financial	/training.html	Offline training progress, loss curves
Benchmark	/benchmark.html	Synthetic noisy benchmark results
Benchmark	/ltsf_benchmark.html	LTSE (Exchange, ETTh) benchmark results
KLA	/kla_kalman.html	Live Kalman parameter (K , A , R) traces
KLA	/complexity.html	Parameter count $\times$ performance heatmap

Table 5.4: Role of each dashboard view.

A context-sensitive help dialog opens on clicking the ? button in the bottom-right corner of every page. Page-to-page navigation happens through the top bar; the browser’s *back* button also works.

Usage patterns

The system supports four typical scenarios. The paper-trading demo (default) runs on the Binance live feed with simulated execution. The backtest mode runs the same pipeline against historical data without a Binance connection via `adapters/testing/backtest.py`, writing results to the console and CSV. The offline training mode ingests a full dataset (by default 3 days of BTCUSDT 5-minute candles) in a single pass and persists the weights to a `*.pt` file. Finally, a *dashboard-only* mode is useful for reviewing a previous run: the Binance feed can be disabled and the dashboard browsed offline.

Runtime logging and shutdown

`stdout` contains normal status messages (`[BOOT]`, `[SIGNAL]`, `[FILL]`, `[DASHBOARD]`), while `stderr` holds warnings and errors (`[WARN]`, `[FATAL ERROR]`). The two streams can be logged separately:

```
nohup python main.py > run_log.txt 2> run_err.txt &
```

The run can be safely interrupted at any time with **Ctrl+C**. The `finally` clause guarantees that (i) the WebSocket connection is closed, (ii) in-memory event buffers are flushed, (iii) the HTTP server closes active connections, and (iv) the final state (open positions, equity, last signal) is written under the `FINAL DASHBOARD SNAPSHOT` header.

Security considerations

The demo runs in *paper-trading* mode by default and does not submit real Binance orders. If a production key is nonetheless placed in the `.env` file, it is recommended to (a) grant only read and Futures trading permissions on the Binance side, and (b) restrict the key by IP whitelist. The dashboard listens on `0.0.0.0:8080` by default; exposing it on a public IP should only happen behind an external reverse proxy (nginx, traefik).

5.8.5 Troubleshooting

ModuleNotFoundError: mamba_ssm

Install the CUDA-compatible packages from `requirements-ml.txt`, or run on CPU:

```
python main.py --device cpu.
```

Binance API 403/401

Check the `.env` keys and IP whitelist settings on the Binance trading interface.

WebSocket connection error

A firewall may block Binance's `wss://` endpoint (port 9443); set up a proxy if necessary.

5.9 Testing

5.9.1 Testing Strategy

The test suite uses **pytest**. Core logic is covered by unit tests; the Binance and aiohttp adapters are replaced with `unittest.mock`, so all tests run deterministically without a network connection.

Test file	Type	Coverage
<code>test_strategy.py</code>	Unit	ThresholdStrategy signal logic, threshold/config/scale checks
<code>test_risk_policy.py</code>	Unit	RiskPolicy blocking paths, MaxQty, MaxNotional, Confidence, SlippageGuard
<code>test_execution.py</code>	Integration	PaperBroker order execution, fill simulation, Stop/Limit orders
<code>test_equity.py</code>	Unit	SimulationWallet equity calculation, PnL, mark-price updates
<code>test_fee_fix.py</code>	Regression	Commission calculation accuracy across a trading cycle
<code>test_fill_fix.py</code>	Regression	Fill-event time/price logic, partial fill handling
<code>test_signal_diagnostics.py</code>	Unit	Signal diagnostic metrics (edge score, hit-rate calculation)
<code>test_binance_feed.py</code>	Mock-based	BinanceFeed WebSocket event processing, reconnect logic

Table 5.5: Summary of test files.

5.9.2 Running the Tests

```
# Run all tests
```

```
pytest tests/ -v
```

```
# Unit tests only, skipping slow integration tests
```

```
pytest tests/ -v -m "not slow"
```

```
# Coverage report
```

```
pytest tests/ --cov=core --cov-report=term-missing
```

5.9.3 Selected Test Cases

- `test_risk_policy.py` verifies that all blocking rules (`MaxQtyRule`, `MaxNotionalRule`, `ConfidenceRule`, `SlippageGuardRule`) correctly reject in-

valid orders, and that when multiple rules are applied together the strictest one prevails. The tests cover approximately 100 random input combinations.

- `test_strategy.py` verifies that `ThresholdStrategy` emits a BUY/SELL signal exactly when the predicted probability exceeds `threshold`, and that `min_edge`, `min_confidence`, and `max_sigma` correctly filter weak signals.
- All tests in `test_equity.py` and `test_fill_fix.py` pass, confirming that PnL calculation and partial-fill handling work correctly in multi-symbol portfolios.
- `test_fee_fix.py` acts as a regression guard so that a previous bug (double-counting commission in the itemised view) does not reappear.

5.9.4 Testing Limitations

There is no end-to-end integration test that checks the full pipeline (from data ingestion to order execution) in a single run. Backtest mode partially fills this gap, but a dedicated simulation integration test suite would improve coverage.

Chapter 6

Conclusion

6.1 Main Findings

KCMamba integrates a differentiable Kalman filtering mechanism into the Mamba SSM. The experiments on three datasets lead to the following conclusions:

1. Going from $\sigma = 0$ to $\sigma = 5$, KCMamba’s average MSE degradation is +1 528%, versus +28 234% for Fair Mamba, an $18.5\times$ difference. In the Exchange/ $s = 16$ setting this becomes five orders of magnitude: +66% vs. +152 350%.
2. The model learned on its own to reduce K as noise increases ($0.706 \rightarrow 0.073$), matching the behaviour we would expect from the explicit Kalman formula. The architectural prior genuinely guides learning.
3. With stride $s = 16$, the Kalman structure acts as an implicit regulariser that partially compensates for the lack of training windows.

6.2 Limitations

KCMamba does not win in every configuration, and the experimental framework contains several simplifications that meaningfully restrict the generalisability of our conclusions.

- At $\sigma = 0$, $s = 16$ Fair Mamba nearly memorises and achieves a much better MSE. This suggests that in noise-free settings the explicit Kalman structure imposes unnecessary overhead: the always-present filtering term causes mild underfitting, and the architecture cannot fully switch the filter off.

- On ETTh1/ $s = 1$ LSTM is the best; its gating mechanism exploits the large data volume. The forget/input gate pair realises a similar adaptive filter, but more flexibly, because it is not tied to the $K = Q/(Q + R)$ structure. This is not a KCMamba failure but illustrates that the utility of the inductive prior decreases in data-rich regimes.
- Only 5 epochs were run, and every configuration used a single seed. This guarantees neither convergence nor statistical significance: the reported results are point estimates without confidence intervals. A publication-grade study would require at least three repetitions and early stopping.
- Only $L = H = 96$ was evaluated. At 336, 720 and 960-step horizons, Mamba’s $O(T)$ complexity may yield a substantive advantage over $O(T^2)$ Transformers, and LSTM’s memory limitations also begin to dominate; the conclusions here thus formally apply only to short horizons.
- The noise model is exclusively isotropic Gaussian ($\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$); in real applications noise is typically heteroscedastic, temporally correlated, sometimes non-Gaussian, and often structural (sensor dropouts, missing values). Robustness measured here does not automatically generalise to those noise forms.
- Scalar Q_t, R_t estimation assumes per-channel independent noise components. Estimating a diagonal or full covariance matrix would yield a richer noise model at the cost of a noticeably larger parameter count.
- Two benchmarks (Weather, ECL) were excluded from this round because of training-time scaling; dimensional sensitivity (7 vs. 21 vs. 321 channels) is therefore not *a priori* substantiated.
- A broader hyperparameter-sensitivity study (hidden dimension, number of heads, layers, kernel size, learning rate) was not performed; the configuration presented is a reasonable default that is not guaranteed to be Pareto-optimal.
- On the trading side (Chapter 5) the system is validated in backtests; live-market effects, slippage variance, outages and partial-fill ordering lie outside the scope of this thesis.

6.3 Future Work

- A systematic sweep over $s \in \{1, 4, 8, 16, 32, 64\}$ would identify the data-scarcity threshold at which the Kalman structure’s regularising effect first emerges.
- At prediction horizons of 336, 720, or 960 steps, Mamba’s $O(T \log T)$ complexity may confer a meaningful advantage over $O(T^2)$ Transformer baselines.
- Replacing simulated Gaussian perturbations with sensor dropouts, missing values, and outliers would bring the evaluation closer to real-world deployment conditions.
- Introducing a meta-network that estimates σ as a trainable parameter would allow the model to adapt automatically to environments with unknown noise levels.
- Replacing the scalar Q_t, R_t with a diagonal Cholesky factorisation would yield richer uncertainty estimates and potentially improve calibration.

6.4 Closing Remarks

KCMamba demonstrates that classical Kalman filtering principles and modern sequence models are not mutually exclusive. The explicit noise model provides a strong inductive bias, particularly where data is scarce or noise is high. The trading system described in Chapter 5 shows that this model is not just an experimental prototype but a solution integrable into a real trading environment.

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